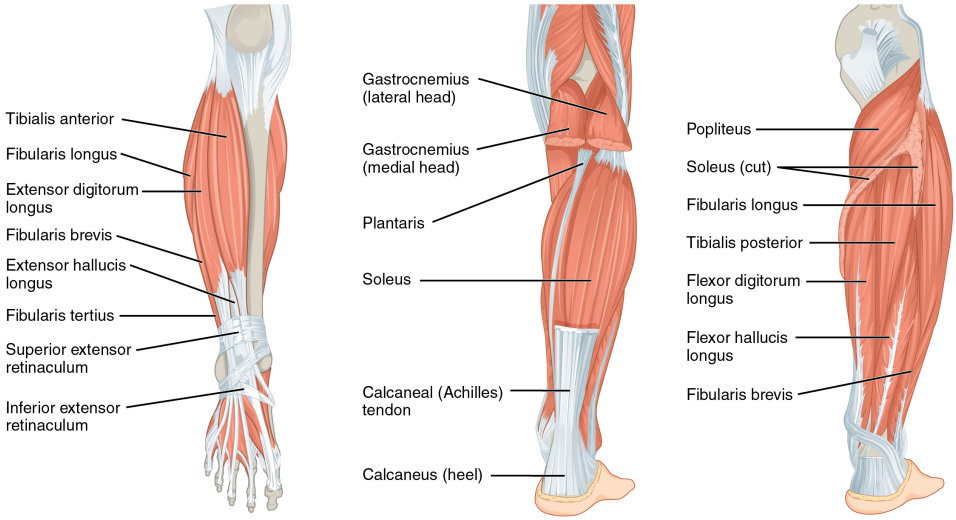
cross-legged.

The **posterior compartment of the thigh** includes muscles that flex the leg and extend the thigh. The three long muscles on the back of the knee are the **hamstring group**, which flexes the knee. These are the **biceps femoris**, **semitendinosus**, and **semimembranosus**. The tendons of these muscles form the **popliteal fossa**, the diamond-shaped space at the back of the knee.

Muscles That Move the Feet and Toes

Similar to the thigh muscles, the muscles of the leg are divided by deep fascia into compartments, although the leg has three: anterior, lateral, and posterior (Figure 11.32 and Figure 11.33).



Superficial muscles of the right lower leg (anterior view)

Superficial muscles of the right lower leg (posterior view)

Deep muscles of the right lower leg (posterior view)

FIGURE 11.32 Muscles of the Lower Leg The muscles of the anterior compartment of the lower leg are generally responsible for dorsiflexion, and the muscles of the posterior compartment of the lower leg are generally responsible for plantar flexion. The lateral and medial muscles in both compartments invert, evert, and rotate the foot.

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Anterior compartment of leg					
Raises the sole of the foot off the ground, as when preparing to foot-tap; bends the inside of the foot upwards, as when catching your balance while falling laterally toward the opposite side as the balancing foot	Foot	Dorsiflexion; inversion	Tibialis anterior	Lateral condyle and upper tibial shaft; interosseous membrane	Interior surface of medial cuneiform; First metatarsal bone
Raises the sole of the foot off the ground, as when preparing to foot-tap; extends the big toe	Foot; big toe	Foot: dorsiflexion; big toe: extension	Extensor hallucis longus	Anteromedial fibula shaft; interosseous membrane	Distal phalanx of big toe
Raises the sole of the foot off the ground, as when preparing to foot-tap; extends toes	Foot; toes 2–5	Foot: dorsiflexion; toes: extension	Extensor digitorum longus	Lateral condyle of tibia; proximal portion of fibula; interosseous membrane	Middle and distal phalanges of toes 2–5
Lateral compartment of leg					
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot downwards, as when catching your balance while falling laterally toward the same side as the balancing foot	Foot	Plantar flexion and eversion	Fibularis longus	Upper portion of lateral fibula	First metatarsal; medial cuneiform
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot downward, as when catching your balance while falling laterally toward the same side as the balancing foot	Foot	Plantar flexion and eversion	Fibularis (peroneus) brevis	Distal fibula shaft	Proximal end of fifth metatarsal
Posterior compartment of leg: Superficial muscles					
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; assists in moving the back of the lower legs up and back toward the buttocks	Foot; tibia/ fibula	Foot: plantar flexion; tibia/fibula: flexion	Gastrocnemius	Medial and lateral condyles of femur	Posterior calcaneus
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; maintains posture while walking	Foot	Plantar flexion	Soleus	Superior tibia; fibula; interosseous membrane	Posterior calcaneus
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; assists in moving the back of the lower legs up and back toward the buttocks	Foot; tibia/ fibula	Foot: plantar flexion; tibia/fibula: flexion	Plantaris	Posterior femur above lateral condyle	Calcaneus or calcaneus tendon
Lowers the sole of the foot to the ground, as when foot-tapping or jumping	Foot	Plantar flexion	Tibialis posterior	Superior tibia and fibula; interosseous membrane	Several tarsals and metatarsals 2–4
Posterior compartment of leg: Deep muscles					
Moves the back of the lower legs up and back toward the buttocks; assists in rotation of the leg at the knee and thigh	Tibia/ fibula	Tibia/fibula: flexion thigh and lower leg: medial and lateral rotation	Popliteus	Lateral condyle of femur; lateral meniscus	Proximal tibia
Lowers the sole of the foot to the ground, as when foot-tapping or jumping; bends the inside of the foot upward and flexes toes	Foot; toes 2–5	Foot: plantar flexion and inversion toes: flexion	Flexor digitorum longus	Posterior tibia	Distal phalanges of toes 2–5
Flexes the big toe	Big toe; foot	Big toe: flexion foot: plantar flexion	Flexor hallucis longus	Midshaft of fibula; interosseous membrane	Distal phalanx of big toe

FIGURE 11.33 Muscles That Move the Feet and Toes

The muscles in the **anterior compartment of the leg**: the **tibialis anterior**, a long and thick muscle on the lateral surface of the tibia, the **extensor hallucis longus**, deep under it, and the **extensor digitorum longus**, lateral to it, all contribute to raising the front of the foot when they contract. The **fibularis tertius**, a small muscle that originates on the anterior surface of the fibula, is associated with the extensor digitorum longus and sometimes fused to it, but is not present in all people. Thick bands of connective tissue called the **superior extensor retinaculum** (transverse ligament of the ankle) and the **inferior extensor retinaculum**, hold the tendons of these muscles in place during dorsiflexion.

The **lateral compartment of the leg** includes two muscles: the **fibularis longus** (peroneus longus) and the **fibularis brevis** (peroneus brevis). The superficial muscles in the **posterior compartment of the leg** all insert onto the **calcaneal tendon** (Achilles tendon), a strong tendon that inserts into the calcaneal bone of the ankle. The muscles in this compartment are large and strong and keep humans upright. The most superficial and visible muscle of the calf is the **gastrocnemius**. Deep to the gastrocnemius is the wide, flat **soleus**. The **plantaris** runs obliquely between the two; some people may have two of these muscles, whereas no plantaris is observed in about seven percent of other cadaver dissections. The plantaris tendon is a desirable substitute for the fascia lata in hernia repair, tendon transplants, and repair of ligaments. There are four deep muscles in the posterior compartment of the leg as well: the **popliteus**, **flexor digitorum longus**, **flexor hallucis longus**, and **tibialis posterior**.

The foot also has intrinsic muscles, which originate and insert within it (similar to the intrinsic muscles of the hand). These muscles primarily provide support for the foot and its arch, and contribute to movements of the toes (Figure 11.34 and Figure 11.35). The principal support for the longitudinal arch of the foot is a deep fascia called **plantar aponeurosis**, which runs from the calcaneus bone to the toes (inflammation of this tissue is the cause of "plantar fasciitis," which can affect runners. The intrinsic muscles of the foot consist of two groups. The **dorsal group** includes only one muscle, the **extensor digitorum brevis**. The second group is the **plantar group**, which consists of four layers, starting with the most superficial.

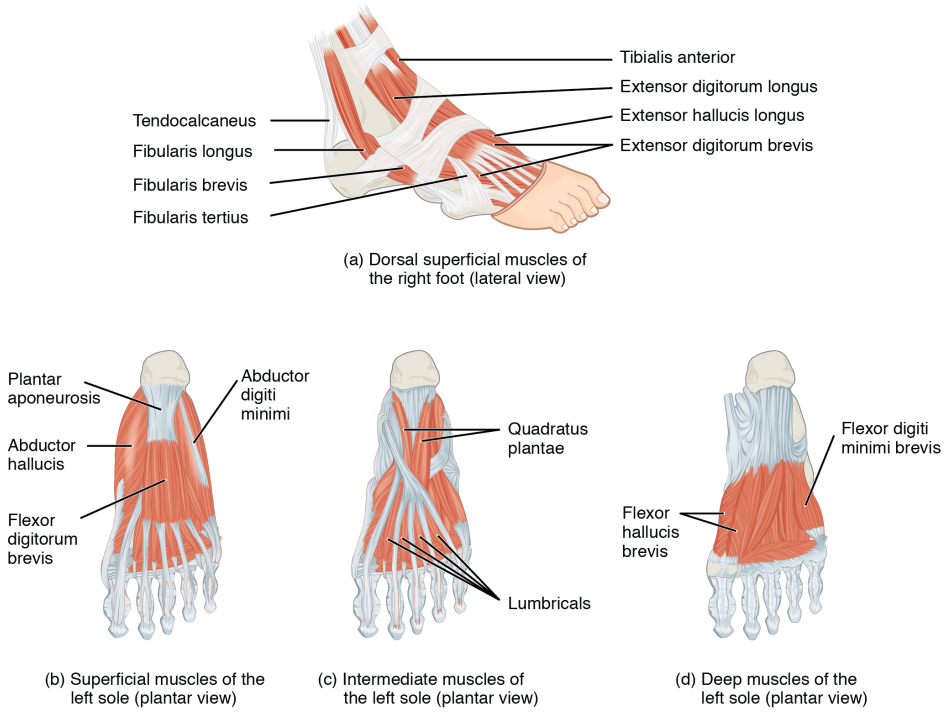


FIGURE 11.34 Intrinsic Muscles of the Foot The muscles along the dorsal side of the foot (a) generally extend the toes while the muscles of the plantar side of the foot (b, c, d) generally flex the toes. The plantar muscles exist in four layers, providing the foot the strength to counterbalance the weight of the body. In this diagram, three of the layers are shown from a plantar view beginning with the bottom-most layer just under the plantar skin of the foot (b) and ending with the top-most layer (d) located just inferior to the foot and toe bones.

Movement	Target	Target motion direction	Prime mover	Origin	Insertion
Dorsal group					
Extends toes 2–5	Toes 2–5	Extension	Extensor digitorum brevis	Calcaneus; extensor retinaculum	Base of proximal phalanx of big toe; extensor expansions on toes 2–5
Plantar group (layer 1)					
Abducts and flexes big toe	Big toe	Adduction; flexion	Abductor hallucis	Calcaneal tuberosity; flexor retinaculum	Proximal phalanx of big toe
Flexes toes 2–4	Middle toes	Flexion	Flexor digitorum brevis	Calcaneal tuberosity	Middle phalanx of toes 2–4
Abducts and flexes small toe	Toe 5	Abduction; flexion	Abductor digiti minimi	Calcaneal tuberosity	Proximal phalanx of little toe
Plantar group (layer 2)					
Assists in flexing toes 2–5	Toes 2–5	Flexion	Quadratus plantae	Medial and lateral sides of calcaneus	Tendon of flexor digitorum longus
Extends toes 2–5 at the interphalangeal joints; flexes the small toes at the metatarsophalangeal joints	Toes 2–5	Extension; flexion	Lumbricals	Tendons of flexor digitorum longus	Medial side of proximal phalanx of toes 2–5
Plantar group (layer 3)					
Flexes big toe	Big toe	Flexion	Flexor hallucis brevis	Lateral cuneiform; cuboid bones	Base of proximal phalanx of big toe
Adducts and flexes big toe	Big toe	Adduction; flexion	Adductor hallucis	Bases of metatarsals 2–4; fibularis longus tendon sheath; ligament across metatarsophalangeal joints	Base of proximal phalanx of big toe
Flexes small toe	Little toe	Flexion	Flexor digiti minimi brevis	Base of metatarsal 5; tendon sheath of fibularis longus	Base of proximal phalanx of little toe
Plantar group (layer 4)					
Abducts and flexes middle toes at metatarsophalangeal joints; extends middle toes at interphalangeal joints	Middle toes	Abduction; flexion; extension	Dorsal interossei	Sides of metatarsals	Both sides of toe 2; for each other toe, extensor expansion over first phalanx on side opposite toe 2
Abducts toes 3–5; flexes proximal phalanges and extends distal phalanges	Small toes	Abduction; flexion; extension	Plantar interossei	Side of each metatarsal that faces metatarsal 2 (absent from metatarsal 2)	Extensor expansion on first phalanx of each toe (except to 2) on side facing toe 2

FIGURE 11.35 Intrinsic Muscles in the Foot

Key Terms

abduct move away from midline in the sagittal plane
abductor moves the bone away from the midline
abductor digiti minimi muscle that abducts the little finger
abductor pollicis brevis muscle that abducts the thumb
abductor pollicis longus muscle that inserts into the first metacarpal
adductor moves the bone toward the midline
adductor brevis muscle that adducts and medially rotates the thigh
adductor longus muscle that adducts, medially rotates, and flexes the thigh
adductor magnus muscle with an anterior fascicle that adducts, medially rotates and flexes the thigh, and a posterior fascicle that assists in thigh extension
adductor pollicis muscle that adducts the thumb
agonist (also, prime mover) muscle whose contraction is responsible for producing a particular motion
anal triangle posterior triangle of the perineum that includes the anus
anconeus small muscle on the lateral posterior elbow that extends the forearm
antagonist muscle that opposes the action of an agonist
anterior compartment of the arm (anterior flexor compartment of the arm) the biceps brachii, brachialis, brachioradialis, and their associated blood vessels and nerves
anterior compartment of the forearm (anterior flexor compartment of the forearm) deep and superficial muscles that originate on the humerus and insert into the hand
anterior compartment of the leg region that includes muscles that dorsiflex the foot
anterior compartment of the thigh region that includes muscles that flex the thigh and extend the leg
anterior scalene a muscle anterior to the middle scalene
appendicular of the arms and legs
axial of the trunk and head
belly bulky central body of a muscle
bi two
biceps brachii two-headed muscle that crosses the shoulder and elbow joints to flex the forearm while assisting in supinating it and flexing the arm at the shoulder
biceps femoris hamstring muscle
bipennate pennate muscle that has fascicles that are located on both sides of the tendon
brachialis muscle deep to the biceps brachii that provides power in flexing the forearm.
brachioradialis muscle that can flex the forearm quickly or help lift a load slowly
brevis short
buccinator muscle that compresses the cheek
calcaneal tendon (also, Achilles tendon) strong tendon that inserts into the calcaneal bone of the ankle
caval opening opening in the diaphragm that allows the inferior vena cava to pass through; foramen for the vena cava
circular (also, sphincter) fascicles that are concentrically arranged around an opening
compressor urethrae deep perineal muscle in females
convergent fascicles that extend over a broad area and converge on a common attachment site
coracobrachialis muscle that flexes and adducts the arm
corrugator supercilii prime mover of the eyebrows
deep anterior compartment flexor pollicis longus, flexor digitorum profundus, and their associated blood vessels and nerves
deep posterior compartment of the forearm (deep posterior extensor compartment of the forearm) the abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus, extensor indicis, and their associated blood vessels and nerves
deep transverse perineal deep perineal muscle in males
deglutition swallowing
deltoid shoulder muscle that abducts the arm as well as flexes and medially rotates it, and extends and laterally rotates it
diaphragm skeletal muscle that separates the thoracic and abdominal cavities and is dome-shaped at rest
digastric muscle that has anterior and posterior bellies and elevates the hyoid bone and larynx when one swallows; it also depresses the mandible
dorsal group region that includes the extensor digitorum brevis
dorsal interossei muscles that abduct and flex the three middle fingers at the metacarpophalangeal joints and extend them at the interphalangeal joints
epicranial aponeurosis (also, galea aponeurosis) flat broad tendon that connects the frontalis and occipitalis
erector spinae group large muscle mass of the back; primary extensor of the vertebral column

extensor muscle that increases the angle at the joint
extensor carpi radialis brevis muscle that extends and abducts the hand at the wrist
extensor carpi ulnaris muscle that extends and adducts the hand
extensor digiti minimi muscle that extends the little finger
extensor digitorum muscle that extends the hand at the wrist and the phalanges
extensor digitorum brevis muscle that extends the toes
extensor digitorum longus muscle that is lateral to the tibialis anterior
extensor hallucis longus muscle that is partly deep to the tibialis anterior and extensor digitorum longus
extensor indicis muscle that inserts onto the tendon of the extensor digitorum of the index finger
extensor pollicis brevis muscle that inserts onto the base of the proximal phalanx of the thumb
extensor pollicis longus muscle that inserts onto the base of the distal phalanx of the thumb
extensor radialis longus muscle that extends and abducts the hand at the wrist
extensor retinaculum band of connective tissue that extends over the dorsal surface of the hand
external intercostal superficial intercostal muscles that raise the rib cage
external oblique superficial abdominal muscle with fascicles that extend inferiorly and medially
extrinsic eye muscles originate outside the eye and insert onto the outer surface of the white of the eye, and create eyeball movement
extrinsic muscles of the hand muscles that move the wrists, hands, and fingers and originate on the arm
fascicle muscle fibers bundled by perimysium into a unit
femoral triangle region formed at the junction between the hip and the leg and includes the pectineus, femoral nerve, femoral artery, femoral vein, and deep inguinal lymph nodes
fibularis brevis (also, peroneus brevis) muscle that plantar flexes the foot at the ankle and everts it at the intertarsal joints
fibularis longus (also, peroneus longus) muscle that plantar flexes the foot at the ankle and everts it at the intertarsal joints
fibularis tertius small muscle that is associated with the extensor digitorum longus
fixator synergist that assists an agonist by preventing or reducing movement at another joint, thereby stabilizing the origin of the agonist
flexion movement that decreases the angle of a joint
flexor muscle that decreases the angle at the joint
flexor carpi radialis muscle that flexes and abducts the hand at the wrist
flexor carpi ulnaris muscle that flexes and adducts the hand at the wrist
flexor digiti minimi brevis muscle that flexes the little finger
flexor digitorum longus muscle that flexes the four small toes
flexor digitorum profundus muscle that flexes the phalanges of the fingers and the hand at the wrist
flexor digitorum superficialis muscle that flexes the hand and the digits
flexor hallucis longus muscle that flexes the big toe
flexor pollicis brevis muscle that flexes the thumb
flexor pollicis longus muscle that flexes the distal phalanx of the thumb
flexor retinaculum band of connective tissue that extends over the palmar surface of the hand
frontalis front part of the occipitofrontalis muscle
fusiform muscle that has fascicles that are spindle-shaped to create large bellies
gastrocnemius most superficial muscle of the calf
genioglossus muscle that originates on the mandible and allows the tongue to move downward and forward
geniohyoid muscle that depresses the mandible, and raises and pulls the hyoid bone anteriorly
gluteal group muscle group that extends, flexes, rotates, adducts, and abducts the femur
gluteus maximus largest of the gluteus muscles that extends the femur
gluteus medius muscle deep to the gluteus maximus that abducts the femur at the hip
gluteus minimus smallest of the gluteal muscles and deep to the gluteus medius
gracilis muscle that adducts the thigh and flexes the leg at the knee
hamstring group three long muscles on the back of the leg
hyoglossus muscle that originates on the hyoid bone to move the tongue downward and flatten it
hypothenar group of muscles on the medial aspect of the palm
hypothenar eminence rounded contour of muscle at the base of the little finger
iliacus muscle that, along with the psoas major, makes up the iliopsoas
iliococcygeus muscle that makes up the levator ani along with the pubococcygeus
iliocostalis cervicis muscle of the iliocostalis group associated with the cervical region
iliocostalis group laterally placed muscles of the

erector spinae

iliocostalis lumborum muscle of the iliocostalis group associated with the lumbar region

iliocostalis thoracis muscle of the iliocostalis group associated with the thoracic region

iliopsoas group muscle group consisting of iliacus and psoas major muscles, that flexes the thigh at the hip, rotates it laterally, and flexes the trunk of the body onto the hip

iliotibial tract muscle that inserts onto the tibia; made up of the gluteus maximus and connective tissues of the tensor fasciae latae

inferior extensor retinaculum cruciate ligament of the ankle

inferior gemellus muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip

infrahyoid muscles anterior neck muscles that are attached to, and inferior to the hyoid bone

infraspinatus muscle that laterally rotates the arm

innermost intercostal the deepest intercostal muscles that draw the ribs together

insertion end of a skeletal muscle that is attached to the structure (usually a bone) that is moved when the muscle contracts

intercostal muscles muscles that span the spaces between the ribs

intermediate group of midpalmar muscles

internal intercostal muscles the intermediate intercostal muscles that draw the ribs together

internal oblique flat, intermediate abdominal muscle with fascicles that run perpendicular to those of the external oblique

intrinsic muscles of the hand muscles that move the wrists, hands, and fingers and originate in the palm

ischiococcygeus muscle that assists the levator ani and pulls the coccyx anteriorly

lateral compartment of the leg region that includes the fibularis (peroneus) longus and the fibularis (peroneus) brevis and their associated blood vessels and nerves

lateral pterygoid muscle that moves the mandible from side to side

lateralis to the outside

latissimus dorsi broad, triangular axial muscle located on the inferior part of the back

levator ani pelvic muscle that resists intra-abdominal pressure and supports the pelvic viscera

linea alba white, fibrous band that runs along the midline of the trunk

longissimus capitis muscle of the longissimus group associated with the head region

longissimus cervicis muscle of the longissimus group associated with the cervical region

longissimus group intermediately placed muscles of the erector spinae

longissimus thoracis muscle of the longissimus group associated with the thoracic region

longus long

lumbrical muscle that flexes each finger at the metacarpophalangeal joints and extend each finger at the interphalangeal joints

masseter main muscle for chewing that elevates the mandible to close the mouth

mastication chewing

maximus largest

medial compartment of the thigh a region that includes the adductor longus, adductor brevis, adductor magnus, pectineus, gracilis, and their associated blood vessels and nerves

medial pterygoid muscle that moves the mandible from side to side

medialis to the inside

medius medium

middle scalene longest scalene muscle, located between the anterior and posterior scalenes

minimus smallest

multifidus muscle of the lumbar region that helps extend and laterally flex the vertebral column

multipennate pennate muscle that has a tendon branching within it

mylohyoid muscle that lifts the hyoid bone and helps press the tongue to the top of the mouth

oblique at an angle

obturator externus muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip

obturator internus muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip

occipitalis posterior part of the occipitofrontalis muscle

occipitofrontalis muscle that makes up the scalp with a frontal belly and an occipital belly

omohyoid muscle that has superior and inferior bellies and depresses the hyoid bone

opponens digiti minimi muscle that brings the little finger across the palm to meet the thumb

opponens pollicis muscle that moves the thumb across the palm to meet another finger

orbicularis oculi circular muscle that closes the eye

orbicularis oris circular muscle that moves the lips

origin end of a skeletal muscle that is attached to another structure (usually a bone) in a fixed position

palatoglossus muscle that originates on the soft palate to elevate the back of the tongue

palmar interossei muscles that abduct and flex each finger at the metacarpophalangeal joints and extend each finger at the interphalangeal joints
palmaris longus muscle that provides weak flexion of the hand at the wrist
parallel fascicles that extend in the same direction as the long axis of the muscle
patellar ligament extension of the quadriceps tendon below the patella
pectineus muscle that abducts and flexes the femur at the hip
pectoral girdle shoulder girdle, made up of the clavicle and scapula
pectoralis major thick, fan-shaped axial muscle that covers much of the superior thorax
pectoralis minor muscle that moves the scapula and assists in inhalation
pelvic diaphragm muscular sheet that comprises the levator ani and the ischiococcygeus
pelvic girdle hips, a foundation for the lower limb
pennate fascicles that are arranged differently based on their angles to the tendon
perineum diamond-shaped region between the pubic symphysis, coccyx, and ischial tuberosities
piriformis muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip
plantar aponeurosis muscle that supports the longitudinal arch of the foot
plantar group four-layered group of intrinsic foot muscles
plantaris muscle that runs obliquely between the gastrocnemius and the soleus
popliteal fossa diamond-shaped space at the back of the knee
popliteus muscle that flexes the leg at the knee and creates the floor of the popliteal fossa
posterior compartment of the leg region that includes the superficial gastrocnemius, soleus, and plantaris, and the deep popliteus, flexor digitorum longus, flexor hallucis longus, and tibialis posterior
posterior compartment of the thigh region that includes muscles that flex the leg and extend the thigh
posterior scalene smallest scalene muscle, located posterior to the middle scalene
prime mover (also, agonist) principle muscle involved in an action
pronator quadratus pronator that originates on the ulna and inserts on the radius
pronator teres pronator that originates on the humerus and inserts on the radius
psoas major muscle that, along with the iliacus, makes up the iliopsoas
pubococcygeus muscle that makes up the levator ani along with the iliococcygeus
quadratus femoris muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip
quadratus lumborum posterior part of the abdominal wall that helps with posture and stabilization of the body
quadriceps femoris group four muscles, that extend and stabilize the knee
quadriceps tendon (also, patellar tendon) tendon common to all four quadriceps muscles, inserts into the patella
rectus straight
rectus abdominis long, linear muscle that extends along the middle of the trunk
rectus femoris quadricep muscle on the anterior aspect of the thigh
rectus sheaths tissue that makes up the linea alba
retinacula fibrous bands that sheath the tendons at the wrist
rhomboid major muscle that attaches the vertebral border of the scapula to the spinous process of the thoracic vertebrae
rhomboid minor muscle that attaches the vertebral border of the scapula to the spinous process of the thoracic vertebrae
rotator cuff (also, musculotendinous cuff) the circle of tendons around the shoulder joint
sartorius band-like muscle that flexes, abducts, and laterally rotates the leg at the hip
scalene muscles flex, laterally flex, and rotate the head; contribute to deep inhalation
segmental muscle group interspinales and intertransversarii muscles that bring together the spinous and transverse processes of each consecutive vertebra
semimembranosus hamstring muscle
semispinalis capitis transversospinales muscle associated with the head region
semispinalis cervicis transversospinales muscle associated with the cervical region
semispinalis thoracis transversospinales muscle associated with the thoracic region
semitendinosus hamstring muscle
serratus anterior large and flat muscle that originates on the ribs and inserts onto the scapula
soleus wide, flat muscle deep to the gastrocnemius
sphincter urethrovaginalis deep perineal muscle in females
spinalis capitis muscle of the spinalis group associated with the head region

spinalis cervicis muscle of the spinalis group associated with the cervical region
spinalis group medially placed muscles of the erector spinae
spinalis thoracis muscle of the spinalis group associated with the thoracic region
splenius posterior neck muscles; includes the splenius capitis and splenius cervicis
splenius capitis neck muscle that inserts into the head region
splenius cervicis neck muscle that inserts into the cervical region
sternocleidomastoid major muscle that laterally flexes and rotates the head
sternohyoid muscle that depresses the hyoid bone
sternothyroid muscle that depresses the larynx's thyroid cartilage
styloglossus muscle that originates on the styloid bone, and allows upward and backward motion of the tongue
stylohyoid muscle that elevates the hyoid bone posteriorly
subclavius muscle that stabilizes the clavicle during movement
subscapularis muscle that originates on the anterior scapula and medially rotates the arm
superficial anterior compartment of the forearm flexor carpi radialis, palmaris longus, flexor carpi ulnaris, flexor digitorum superficialis, and their associated blood vessels and nerves
superficial posterior compartment of the forearm extensor radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, extensor carpi ulnaris, and their associated blood vessels and nerves
superior extensor retinaculum transverse ligament of the ankle
superior gemellus muscle deep to the gluteus maximus on the lateral surface of the thigh that laterally rotates the femur at the hip
supinator muscle that moves the palm and forearm anteriorly
suprahyoid muscles neck muscles that are superior to the hyoid bone
supraspinatus muscle that abducts the arm
synergist muscle whose contraction helps a prime mover in an action
temporalis muscle that retracts the mandible
tendinous intersections three transverse bands of collagen fibers that divide the rectus abdominis into segments
tensor fascia lata muscle that flexes and abducts the thigh
teres major muscle that extends the arm and assists in adduction and medial rotation of it
teres minor muscle that laterally rotates and extends the arm
thenar group of muscles on the lateral aspect of the palm
thenar eminence rounded contour of muscle at the base of the thumb
thyrohyoid muscle that depresses the hyoid bone and elevates the larynx's thyroid cartilage
tibialis anterior muscle located on the lateral surface of the tibia
tibialis posterior muscle that plantar flexes and inverts the foot
transversospinales muscles that originate at the transverse processes and insert at the spinous processes of the vertebrae
transversus abdominis deep layer of the abdomen that has fascicles arranged transversely around the abdomen
trapezius muscle that stabilizes the upper part of the back
tri three
triceps brachii three-headed muscle that extends the forearm
unipennate pennate muscle that has fascicles located on one side of the tendon
urogenital triangle anterior triangle of the perineum that includes the external genitals
vastus intermedius quadricep muscle that is between the vastus lateralis and vastus medialis and is deep to the rectus femoris
vastus lateralis quadricep muscle on the lateral aspect of the thigh
vastus medialis quadricep muscle on the medial aspect of the thigh

Chapter Review

11.1 Interactions of Skeletal Muscles, Their Fascicle Arrangement, and Their Lever Systems

Skeletal muscles each have an origin and an insertion. The end of the muscle that attaches to the bone being pulled is called the muscle's insertion and the end of the muscle attached to a fixed, or stabilized, bone is called the origin. The muscle primarily responsible for a movement is called the prime mover, and muscles that assist in this action are called synergists. A synergist that makes the insertion site more stable is

called a fixator. Meanwhile, a muscle with the opposite action of the prime mover is called an antagonist. Several factors contribute to the force generated by a skeletal muscle. One is the arrangement of the fascicles in the skeletal muscle. Fascicles can be parallel, circular, convergent, pennate, fusiform, or triangular. Each arrangement has its own range of motion and ability to do work.

11.2 Naming Skeletal Muscles

Muscle names are based on many characteristics. The location of a muscle in the body is important. Some muscles are named based on their size and location, such as the gluteal muscles of the buttocks. Other muscle names can indicate the location in the body or bones with which the muscle is associated, such as the tibialis anterior. The shapes of some muscles are distinctive; for example, the direction of the muscle fibers is used to describe muscles of the body midline. The origin and/or insertion can also be features used to name a muscle; examples are the biceps brachii, triceps brachii, and the pectoralis major.

11.3 Axial Muscles of the Head, Neck, and Back

Muscles are either axial muscles or appendicular. The axial muscles are grouped based on location, function, or both. Some axial muscles cross over to the appendicular skeleton. The muscles of the head and neck are all axial. The muscles in the face create facial expression by inserting into the skin rather than onto bone. Muscles that move the eyeballs are extrinsic, meaning they originate outside of the eye and insert onto it. Tongue muscles are both extrinsic and intrinsic. The genioglossus depresses the tongue and moves it anteriorly; the styloglossus lifts the tongue and retracts it; the palatoglossus elevates the back of the tongue; and the hyoglossus depresses and flattens it. The muscles of the anterior neck facilitate swallowing and speech, stabilize the hyoid bone and position the larynx. The muscles of the neck stabilize and move the head. The sternocleidomastoid divides the neck into anterior and posterior triangles.

The muscles of the back and neck that move the vertebral column are complex, overlapping, and can be divided into five groups. The splenius group includes the splenius capitis and the splenius cervicis. The erector spinae has three subgroups. The iliocostalis group includes the iliocostalis cervicis, the iliocostalis thoracis, and the iliocostalis lumborum. The longissimus group includes the longissimus capitis, the longissimus cervicis, and the longissimus thoracis. The spinalis group includes the spinalis capitis, the spinalis cervicis, and the spinalis thoracis. The transversospinales include the semispinalis capitis, semispinalis cervicis, semispinalis thoracis, multifidus, and rotatores. The segmental muscles include the interspinales and intertransversarii. Finally, the scalenes include the anterior scalene, middle scalene, and posterior scalene.

11.4 Axial Muscles of the Abdominal Wall, and Thorax

Made of skin, fascia, and four pairs of muscle, the anterior abdominal wall protects the organs located in the abdomen and moves the vertebral column. These muscles include the rectus abdominis, which extends through the entire length of the trunk, the external oblique, the internal oblique, and the transversus abdominis. The quadratus lumborum forms the posterior abdominal wall.

The muscles of the thorax play a large role in breathing, especially the dome-shaped diaphragm. When it contracts and flattens, the volume inside the pleural cavities increases, which decreases the pressure within them. As a result, air will flow into the lungs. The external and internal intercostal muscles span the space between the ribs and help change the shape of the rib cage and the volume-pressure ratio inside the pleural cavities during inspiration and expiration.

The perineum muscles play roles in urination in both sexes, ejaculation in males, and vaginal contraction in females. The pelvic floor muscles support the pelvic organs, resist intra-abdominal pressure, and work as sphincters for the urethra, rectum, and vagina.

11.5 Muscles of the Pectoral Girdle and Upper Limbs

The clavicle and scapula make up the pectoral girdle, which provides a stable origin for the muscles that move the humerus. The muscles that position and stabilize the pectoral girdle are located on the thorax. The anterior thoracic muscles are the subclavius, pectoralis minor, and the serratus anterior. The posterior thoracic muscles are the trapezius, levator scapulae, rhomboid major, and rhomboid minor. Nine muscles cross the shoulder joint to move the humerus. The ones that originate on the axial skeleton are the pectoralis major and the latissimus dorsi. The deltoid, subscapularis, supraspinatus, infraspinatus, teres major, teres minor, and coracobrachialis originate on the scapula.

The forearm flexors include the biceps brachii, brachialis, and brachioradialis. The extensors are the

triceps brachii and anconeus. The pronators are the pronator teres and the pronator quadratus. The supinator is the only one that turns the forearm anteriorly.

The extrinsic muscles of the hands originate along the forearm and insert into the hand in order to facilitate crude movements of the wrists, hands, and fingers. The superficial anterior compartment of the forearm produces flexion. These muscles are the flexor carpi radialis, palmaris longus, flexor carpi ulnaris, and the flexor digitorum superficialis. The deep anterior compartment produces flexion as well. These are the flexor pollicis longus and the flexor digitorum profundus. The rest of the compartments produce extension. The extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, and extensor carpi ulnaris are the muscles found in the superficial posterior compartment. The deep posterior compartment includes the abductor longus, extensor pollicis brevis, extensor pollicis longus, and the extensor indicis.

Finally, the intrinsic muscles of the hands allow our fingers to make precise movements, such as typing and writing. They both originate and insert within the hand. The thenar muscles, which are located on the lateral part of the palm, are the abductor pollicis brevis, opponens pollicis, flexor pollicis brevis, and adductor pollicis. The hypothenar muscles, which are located on the medial part of the palm, are the abductor digiti minimi, flexor digiti minimi brevis, and opponens digiti minimi. The intermediate muscles, located in the middle of the palm, are the lumbricals, palmar interossei, and dorsal interossei.

11.6 Appendicular Muscles of the Pelvic Girdle and Lower Limbs

The pelvic girdle attaches the legs to the axial skeleton. The hip joint is where the pelvic girdle and the leg come together. The hip is joined to the pelvic girdle by many muscles. In the gluteal region, the psoas major and iliacus form the iliopsoas. The large and strong gluteus maximus, gluteus medius, and gluteus minimus extend and abduct the femur. Along with the gluteus maximus, the tensor fascia lata muscle forms the iliotibial tract. The lateral rotators of the femur at the hip are the piriformis, obturator internus, obturator externus, superior gemellus, inferior gemellus, and quadratus femoris. On the medial part of the thigh, the adductor longus, adductor brevis, and adductor magnus adduct the thigh and medially rotate it. The pectineus muscle adducts and flexes the femur at the hip.

The thigh muscles that move the femur, tibia, and fibula are divided into medial, anterior, and posterior compartments. The medial compartment includes the adductors, pectineus, and the gracilis. The anterior compartment comprises the quadriceps femoris, quadriceps tendon, patellar ligament, and the sartorius. The quadriceps femoris is made of four muscles: the rectus femoris, the vastus lateralis, the vastus medius, and the vastus intermedius, which together extend the knee. The posterior compartment of the thigh includes the hamstrings: the biceps femoris, semitendinosus, and the semimembranosus, which all flex the knee.

The muscles of the leg that move the foot and toes are divided into anterior, lateral, superficial- and deep-posterior compartments. The anterior compartment includes the tibialis anterior, the extensor hallucis longus, the extensor digitorum longus, and the fibularis (peroneus) tertius. The lateral compartment houses the fibularis (peroneus) longus and the fibularis (peroneus) brevis. The superficial posterior compartment has the gastrocnemius, soleus, and plantaris; and the deep posterior compartment has the popliteus, tibialis posterior, flexor digitorum longus, and flexor hallucis longus.

Review Questions

1. Which of the following is unique to the muscles of facial expression?
   a. They all originate from the scalp musculature.
   b. They insert onto the cartilage found around the face.
   c. They only insert onto the facial bones.
   d. They insert into the skin.

2. Which of the following helps an agonist work?
   a. a synergist
   b. a fixator
   c. an insertion
   d. an antagonist

3. Which of the following statements is correct about what happens during flexion?
   a. The angle between bones is increased.
   b. The angle between bones is decreased.
   c. The bone moves away from the body.
   d. The bone moves toward the center of the body.

4. Which is moved the *least* during muscle contraction?
   a. the origin
   b. the insertion
   c. the ligaments
   d. the joints

5. Which muscle has a convergent pattern of fascicles?
   a. biceps brachii
   b. gluteus maximus
   c. pectoralis major
   d. rectus femoris

6. A muscle that has a pattern of fascicles running along the long axis of the muscle has which of the following fascicle arrangements?
   a. circular
   b. pennate
   c. parallel
   d. rectus

7. Which arrangement *best* describes a bipennate muscle?
   a. The muscle fibers feed in on an angle to a long tendon from both sides.
   b. The muscle fibers feed in on an angle to a long tendon from all directions.
   c. The muscle fibers feed in on an angle to a long tendon from one side.
   d. The muscle fibers on one side of a tendon feed into it at a certain angle and muscle fibers on the other side of the tendon feed into it at the opposite angle.

8. The location of a muscle's insertion and origin can determine _____.
   a. action
   b. the force of contraction
   c. muscle name
   d. the load a muscle can carry

9. Where is the temporalis muscle located?
   a. on the forehead
   b. in the neck
   c. on the side of the head
   d. on the chin

10. Which muscle name does *not* make sense?
   a. extensor digitorum
   b. gluteus minimus
   c. biceps femoris
   d. extensor minimus longus

11. Which of the following terms would be used in the name of a muscle that moves the leg away from the body?
   a. flexor
   b. adductor

c. extensor
d. abductor

12. Which of the following is a prime mover in head flexion?
a. occipitofrontalis
b. corrugator supercilii
c. sternocleidomastoid
d. masseter

13. Where is the inferior oblique muscle located?
a. in the abdomen
b. in the eye socket
c. in the anterior neck
d. in the face

14. What is the action of the masseter?
a. swallowing
b. chewing
c. moving the lips
d. closing the eye

15. The names of the extrinsic tongue muscles commonly end in _____.
a. -glottis
b. -glossus
c. -gluteus
d. -hyoid

16. What is the function of the erector spinae?
a. movement of the arms
b. stabilization of the pelvic girdle
c. postural support
d. rotating of the vertebral column

17. Which of the following abdominal muscles is not a part of the anterior abdominal wall?
a. quadratus lumborum
b. rectus abdominis
c. interior oblique
d. exterior oblique

18. Which muscle pair plays a role in respiration?
a. intertransversarii, interspinales
b. semispinalis cervicis, semispinalis thoracis
c. trapezius, rhomboids
d. diaphragm, scalene

19. What is the linea alba?
a. a small muscle that helps with compression of the abdominal organs
b. a long tendon that runs down the middle of the rectus abdominis
c. a long band of collagen fibers that connects the hip to the knee
d. another name for the tendinous inscription

20. The rhomboid major and minor muscles are deep to the _____.
a. rectus abdominis

b. scalene muscles
c. trapezius
d. ligamentum nuchae

21. Which muscle extends the forearm?
a. biceps brachii
b. triceps brachii
c. brachialis
d. deltoid

22. What is the origin of the wrist flexors?
a. the lateral epicondyle of the humerus
b. the medial epicondyle of the humerus
c. the carpal bones of the wrist
d. the deltoid tuberosity of the humerus

23. Which muscles stabilize the pectoral girdle?
a. axial and scapular
b. axial
c. appendicular
d. axial and appendicular

24. The large muscle group that attaches the leg to the pelvic girdle and produces extension of the hip joint is the _____ group.
a. gluteal
b. obturator
c. adductor
d. abductor

25. Which muscle produces movement that allows you to cross your legs?
a. the gluteus maximus
b. the piriformis
c. the gracilis
d. the sartorius

26. What is the largest muscle in the lower leg?
a. soleus
b. gastrocnemius
c. tibialis anterior
d. tibialis posterior

27. The vastus intermedius muscle is deep to which of the following muscles?
a. biceps femoris
b. rectus femoris
c. vastus medialis
d. vastus lateralis

Critical Thinking Questions

28. What effect does fascicle arrangement have on a muscle's action?

29. Movements of the body occur at joints. Describe how muscles are arranged around the joints of the body.

30. Explain how a synergist assists an agonist by being a fixator.

31. Describe the different criteria that contribute to

how skeletal muscles are named.

32. Explain the difference between axial and appendicular muscles.

33. Describe the muscles of the anterior neck.

34. Why are the muscles of the face different from typical skeletal muscle?

35. Describe the fascicle arrangement in the muscles of the abdominal wall. How do they relate to each other?

36. What are some similarities and differences between the diaphragm and the pelvic diaphragm?

37. The tendons of which muscles form the rotator cuff? Why is the rotator cuff important?

38. List the general muscle groups of the shoulders and upper limbs as well as their subgroups.

39. Which muscles form the hamstrings? How do they function together?

40. Which muscles form the quadriceps? How do they function together?

CHAPTER 12

The Nervous System and Nervous Tissue

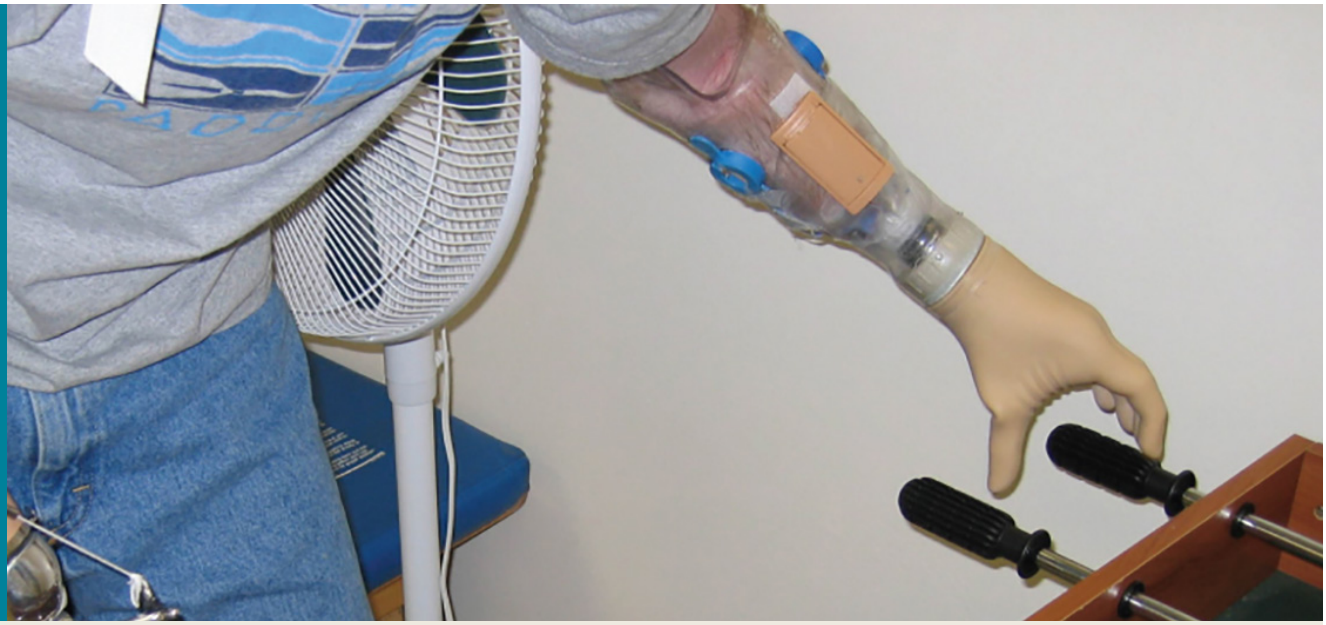


FIGURE 12.1 Robotic Arms Playing Foosball As the neural circuitry of the nervous system has become more fully understood and robotics more sophisticated, it is now possible to integrate technology with the body and restore abilities following traumatic events. At some point in the future, will this type of technology lead to the ability to augment our nervous systems? (credit: U.S. Army/Wikimedia Commons)

Chapter Objectives

After studying this chapter, you will be able to:

- Name the major divisions of the nervous system, both anatomical and functional
- Describe the functional and structural differences between gray matter and white matter structures
- Name the parts of the multipolar neuron in order of polarity
- List the types of glial cells and assign each to the proper division of the nervous system, along with their function(s)
- Distinguish the major functions of the nervous system: sensation, integration, and response
- Describe the components of the membrane that establish the resting membrane potential
- Describe the changes that occur to the membrane that result in the action potential
- Explain the differences between types of graded potentials
- Categorize the major neurotransmitters by chemical type and effect

INTRODUCTION The nervous system is a very complex organ system. In Peter D. Kramer's book *Listening to Prozac*, a pharmaceutical researcher is quoted as saying, "If the human brain were simple enough for us to understand, we would be too simple to understand it" (1994). That quote is from the early 1990s; in the two decades since, progress has continued at an amazing rate within the scientific disciplines of neuroscience. It is an interesting conundrum to consider that the complexity of the nervous system may be too complex for it (that is, for us) to completely unravel. But our current level of understanding is probably nowhere close to that limit.

One easy way to begin to understand the structure of the nervous system is to start with the large divisions and work through to a more in-depth understanding. In other chapters, the finer details of the nervous system will be explained, but first looking at an overview of the system will allow you to begin to understand how its parts work together. The focus of this chapter is on nervous (neural) tissue, both its structure and its function. But before you learn about that, you will see a big picture of the system—actually, a few big pictures.

12.1 Basic Structure and Function of the Nervous System

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify the anatomical and functional divisions of the nervous system
- Relate the functional and structural differences between gray matter and white matter structures of the nervous system to the structure of neurons
- List the basic functions of the nervous system

The picture you have in your mind of the nervous system probably includes the **brain**, the nervous tissue contained within the cranium, and the **spinal cord**, the extension of nervous tissue within the vertebral column. That suggests it is made of two organs—and you may not even think of the spinal cord as an organ—but the nervous system is a very complex structure. Within the brain, many different and separate regions are responsible for many different and separate functions. It is as if the nervous system is composed of many organs that all look similar and can only be differentiated using tools such as the microscope or electrophysiology. In comparison, it is easy to see that the stomach is different than the esophagus or the liver, so you can imagine the digestive system as a collection of specific organs.

The Central and Peripheral Nervous Systems

The nervous system can be divided into two major regions: the central and peripheral nervous systems. The **central nervous system (CNS)** is the brain and spinal cord, and the **peripheral nervous system (PNS)** is everything else (Figure 12.2). The brain is contained within the cranial cavity of the skull, and the spinal cord is contained within the vertebral cavity of the vertebral column. It is a bit of an oversimplification to say that the CNS is what is inside these two cavities and the peripheral nervous system is outside of them, but that is one way to start to think about it. In actuality, there are some elements of the peripheral nervous system that are within the cranial or vertebral cavities. The peripheral nervous system is so named because it is on the periphery—meaning beyond the brain and spinal cord. Depending on different aspects of the nervous system, the dividing line between central and peripheral is not necessarily universal.

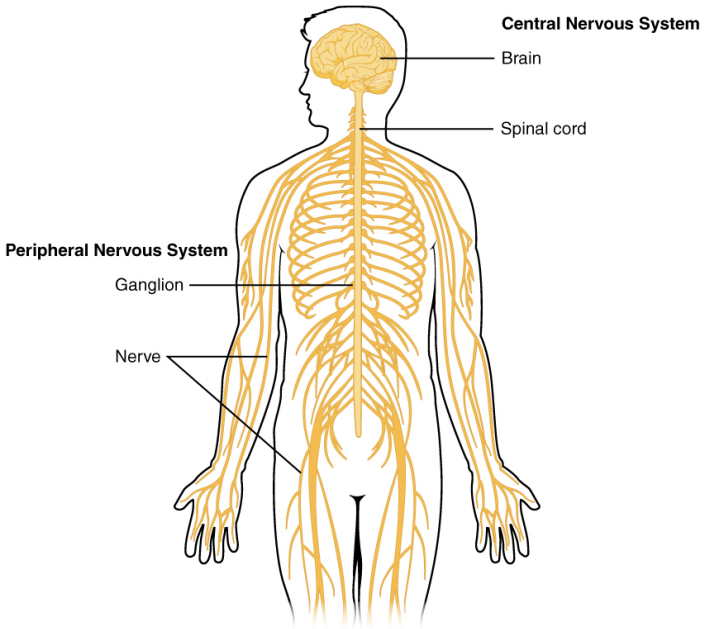


FIGURE 12.2 Central and Peripheral Nervous System The structures of the PNS are referred to as ganglia and nerves, which can be seen as distinct structures. The equivalent structures in the CNS are not obvious from this overall perspective and are best examined in prepared tissue under the microscope.

Nervous tissue, present in both the CNS and PNS, contains two basic types of cells: neurons and glial cells. A **glial cell** is one of a variety of cells that provide a framework of tissue that supports the neurons and their activities. The

neuron is the more functionally important of the two, in terms of the communicative function of the nervous system. To describe the functional divisions of the nervous system, it is important to understand the structure of a neuron. Neurons are cells and therefore have a **soma**, or cell body, but they also have extensions of the cell; each extension is generally referred to as a **process**. There is one important process that every neuron has called an **axon**, which is the fiber that connects a neuron with its target. Another type of process that branches off from the soma is the **dendrite**. Dendrites are responsible for receiving most of the input from other neurons. Looking at nervous tissue, there are regions that predominantly contain cell bodies and regions that are largely composed of just axons. These two regions within nervous system structures are often referred to as **gray matter** (the regions with many cell bodies and dendrites) or **white matter** (the regions with many axons). Figure 12.3 demonstrates the appearance of these regions in the brain and spinal cord. The colors ascribed to these regions are what would be seen in "fresh," or unstained, nervous tissue. Gray matter is not necessarily gray. It can be pinkish because of blood content, or even slightly tan, depending on how long the tissue has been preserved. But white matter is white because axons are insulated by a lipid-rich substance called **myelin**. Lipids can appear as white ("fatty") material, much like the fat on a raw piece of chicken or beef. Actually, gray matter may have that color ascribed to it because next to the white matter, it is just darker—hence, gray.

The distinction between gray matter and white matter is most often applied to central nervous tissue, which has large regions that can be seen with the unaided eye. When looking at peripheral structures, often a microscope is used and the tissue is stained with artificial colors. That is not to say that central nervous tissue cannot be stained and viewed under a microscope, but unstained tissue is most likely from the CNS—for example, a frontal section of the brain or cross section of the spinal cord.

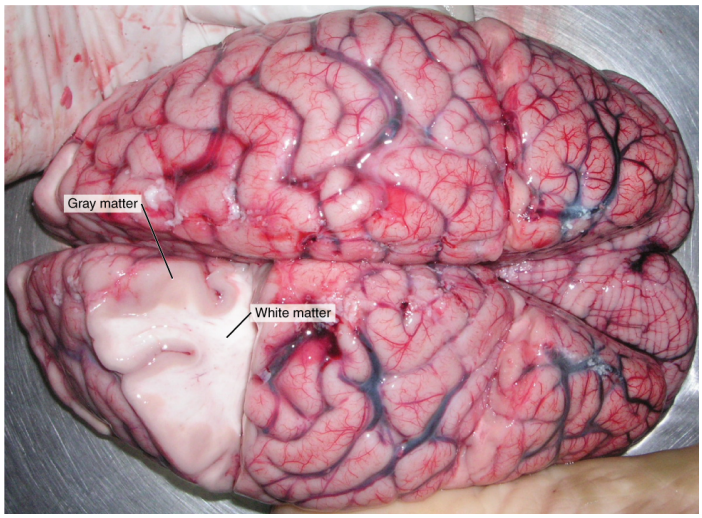


FIGURE 12.3 Gray Matter and White Matter A brain removed during an autopsy, with a partial section removed, shows white matter surrounded by gray matter. Gray matter makes up the outer cortex of the brain. (credit: modification of work by "Suseno"/Wikimedia Commons)

Regardless of the appearance of stained or unstained tissue, the cell bodies of neurons or axons can be located in discrete anatomical structures that need to be named. Those names are specific to whether the structure is central or peripheral. A localized collection of neuron cell bodies in the CNS is referred to as a **nucleus**. In the PNS, a cluster of neuron cell bodies is referred to as a **ganglion**. Figure 12.4 indicates how the term nucleus has a few different meanings within anatomy and physiology. It is the center of an atom, where protons and neutrons are found; it is the center of a cell, where the DNA is found; and it is a center of some function in the CNS. There is also a potentially confusing use of the word ganglion (plural = ganglia) that has a historical explanation. In the central nervous system, there is a group of nuclei that are connected together and were once called the basal ganglia before "ganglion" became accepted as a description for a peripheral structure. Some sources refer to this group of nuclei as the "basal nuclei" to avoid confusion.

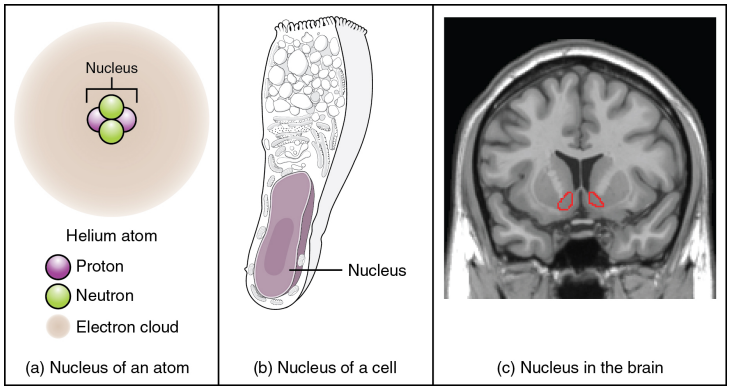


FIGURE 12.4 What Is a Nucleus? (a) The nucleus of an atom contains its protons and neutrons. (b) The nucleus of a cell is the organelle that contains DNA. (c) A nucleus in the CNS is a localized center of function with the cell bodies of several neurons, shown here circled in red. (credit c: "Was a bee"/Wikimedia Commons)

Terminology applied to bundles of axons also differs depending on location. A bundle of axons, or fibers, found in the CNS is called a **tract** whereas the same thing in the PNS would be called a **nerve**. There is an important point to make about these terms, which is that they can both be used to refer to the same bundle of axons. When those axons are in the PNS, the term is nerve, but if they are CNS, the term is tract. The most obvious example of this is the axons that project from the retina into the brain. Those axons are called the optic nerve as they leave the eye, but when they are inside the cranium, they are referred to as the optic tract. There is a specific place where the name changes, which is the optic chiasm, but they are still the same axons (Figure 12.5). A similar situation outside of science can be described for some roads. Imagine a road called "Broad Street" in a town called "Anyville." The road leaves Anyville and goes to the next town over, called "Hometown." When the road crosses the line between the two towns and is in Hometown, its name changes to "Main Street." That is the idea behind the naming of the retinal axons. In the PNS, they are called the optic nerve, and in the CNS, they are the optic tract. Table 12.1 helps to clarify which of these terms apply to the central or peripheral nervous systems.

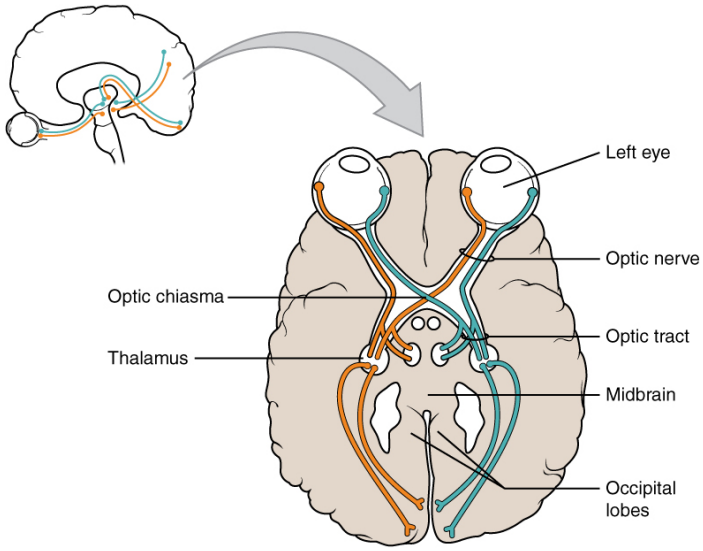


FIGURE 12.5 Optic Nerve Versus Optic Tract This drawing of the connections of the eye to the brain shows the optic nerve extending from the eye to the chiasm, where the structure continues as the optic tract. The same axons extend from the eye to the brain through these two bundles of fibers, but the chiasm represents the border between peripheral and central.

INTERACTIVE LINK

In 2003, the Nobel Prize in Physiology or Medicine was awarded to Paul C. Lauterbur and Sir Peter Mansfield for discoveries related to magnetic resonance imaging (MRI). This is a tool to see the structures of the body (not just the

nervous system) that depends on magnetic fields associated with certain atomic nuclei. The utility of this technique in the nervous system is that fat tissue and water appear as different shades between black and white. Because white matter is fatty (from myelin) and gray matter is not, they can be easily distinguished in MRI images. Try this PhET simulation (http://openstax.org/l/nobel_2) that demonstrates the use of this technology and compares it with other types of imaging technologies. Also, the results from an MRI session are compared with images obtained from X-ray or computed tomography. How do the imaging techniques shown in this game indicate the separation of white and gray matter compared with the freshly dissected tissue shown earlier?

Structures of the CNS and PNS

	CNS	PNS
Group of Neuron Cell Bodies (i.e., gray matter)	Nucleus	Ganglion
Bundle of Axons (i.e., white matter)	Tract	Nerve

TABLE 12.1

Functional Divisions of the Nervous System

The nervous system can also be divided on the basis of its functions, but anatomical divisions and functional divisions are different. The CNS and the PNS both contribute to the same functions, but those functions can be attributed to different regions of the brain (such as the cerebral cortex or the hypothalamus) or to different ganglia in the periphery. The problem with trying to fit functional differences into anatomical divisions is that sometimes the same structure can be part of several functions. For example, the optic nerve carries signals from the retina that are either used for the conscious perception of visual stimuli, which takes place in the cerebral cortex, or for the reflexive responses of smooth muscle tissue that are processed through the hypothalamus.

There are two ways to consider how the nervous system is divided functionally. First, the basic functions of the nervous system are sensation, integration, and response. Secondly, control of the body can be somatic or autonomic—divisions that are largely defined by the structures that are involved in the response. There is also a region of the peripheral nervous system that is called the enteric nervous system that is responsible for a specific set of the functions within the realm of autonomic control related to gastrointestinal functions.

Basic Functions

The nervous system is involved in receiving information about the environment around us (sensation) and generating responses to that information (motor responses). The nervous system can be divided into regions that are responsible for **sensation** (sensory functions) and for the **response** (motor functions). But there is a third function that needs to be included. Sensory input needs to be integrated with other sensations, as well as with memories, emotional state, or learning (cognition). Some regions of the nervous system are termed **integration** or association areas. The process of integration combines sensory perceptions and higher cognitive functions such as memories, learning, and emotion to produce a response.

Sensation. The first major function of the nervous system is sensation—receiving information about the environment to gain input about what is happening outside the body (or, sometimes, within the body). The sensory functions of the nervous system register the presence of a change from homeostasis or a particular event in the environment, known as a **stimulus**. The senses we think of most are the "big five": taste, smell, touch, sight, and hearing. The stimuli for taste and smell are both chemical substances (molecules, compounds, ions, etc.), touch is physical or mechanical stimuli that interact with the skin, sight is light stimuli, and hearing is the perception of sound, which is a physical stimulus similar to some aspects of touch. There are actually more senses than just those, but that list represents the major senses. Those five are all senses that receive stimuli from the outside world, and of which there is conscious perception. Additional sensory stimuli might be from the internal environment (inside the body), such as the stretch of an organ wall or the concentration of certain ions in the blood.

Response. The nervous system produces a response on the basis of the stimuli perceived by sensory structures. An

obvious response would be the movement of muscles, such as withdrawing a hand from a hot stove, but there are broader uses of the term. The nervous system can cause the contraction of all three types of muscle tissue. For example, skeletal muscle contracts to move the skeleton, cardiac muscle is influenced as heart rate increases during exercise, and smooth muscle contracts as the digestive system moves food along the digestive tract. Responses also include the neural control of glands in the body as well, such as the production and secretion of sweat by the eccrine and merocrine sweat glands found in the skin to lower body temperature.

Responses can be divided into those that are voluntary or conscious (contraction of skeletal muscle) and those that are involuntary (contraction of smooth muscles, regulation of cardiac muscle, activation of glands). Voluntary responses are governed by the somatic nervous system and involuntary responses are governed by the autonomic nervous system, which are discussed in the next section.

Integration. Stimuli that are received by sensory structures are communicated to the nervous system where that information is processed. This is called integration. Stimuli are compared with, or integrated with, other stimuli, memories of previous stimuli, or the state of a person at a particular time. This leads to the specific response that will be generated. Seeing a baseball pitched to a batter will not automatically cause the batter to swing. The trajectory of the ball and its speed will need to be considered. Maybe the count is three balls and one strike, and the batter wants to let this pitch go by in the hope of getting a walk to first base. Or maybe the batter's team is so far ahead, it would be fun to just swing away.

Controlling the Body

The nervous system can be divided into two parts mostly on the basis of a functional difference in responses. The **somatic nervous system (SNS)** is responsible for conscious perception and voluntary motor responses. Voluntary motor response means the contraction of skeletal muscle, but those contractions are not always voluntary in the sense that you have to want to perform them. Some somatic motor responses are reflexes, and often happen without a conscious decision to perform them. If your friend jumps out from behind a corner and yells "Boo!" you will be startled and you might scream or leap back. You didn't decide to do that, and you may not have wanted to give your friend a reason to laugh at your expense, but it is a reflex involving skeletal muscle contractions. Other motor responses become automatic (in other words, unconscious) as a person learns motor skills (referred to as "habit learning" or "procedural memory").

The **autonomic nervous system (ANS)** is responsible for involuntary control of the body, usually for the sake of homeostasis (regulation of the internal environment). Sensory input for autonomic functions can be from sensory structures tuned to external or internal environmental stimuli. The motor output extends to smooth and cardiac muscle as well as glandular tissue. The role of the autonomic system is to regulate the organ systems of the body, which usually means to control homeostasis. Sweat glands, for example, are controlled by the autonomic system. When you are hot, sweating helps cool your body down. That is a homeostatic mechanism. But when you are nervous, you might start sweating also. That is not homeostatic, it is the physiological response to an emotional state.

There is another division of the nervous system that describes functional responses. The **enteric nervous system (ENS)** is responsible for controlling the smooth muscle and glandular tissue in your digestive system. It is a large part of the PNS, and is not dependent on the CNS. It is sometimes valid, however, to consider the enteric system to be a part of the autonomic system because the neural structures that make up the enteric system are a component of the autonomic output that regulates digestion. There are some differences between the two, but for our purposes here there will be a good bit of overlap. See Figure 12.6 for examples of where these divisions of the nervous system can be found.

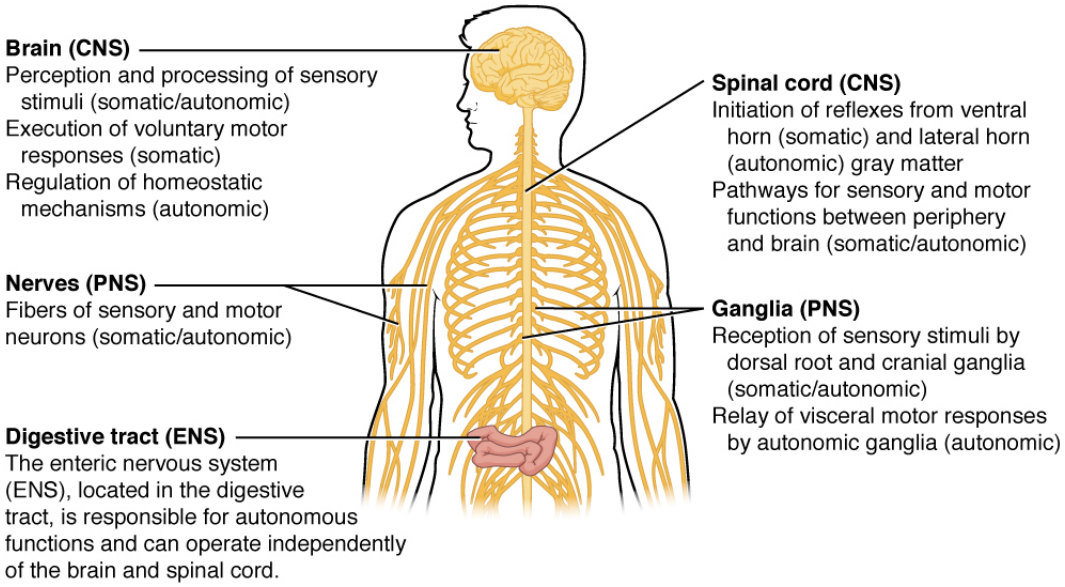


FIGURE 12.6 Somatic, Autonomic, and Enteric Structures of the Nervous System Somatic structures include the spinal nerves, both motor and sensory fibers, as well as the sensory ganglia (posterior root ganglia and cranial nerve ganglia). Autonomic structures are found in the nerves also, but include the sympathetic and parasympathetic ganglia. The enteric nervous system includes the nervous tissue within the organs of the digestive tract.

INTERACTIVE LINK

Visit this site (http://openstax.org/l/troublewstairs) to read about a woman that notices that her daughter is having trouble walking up the stairs. This leads to the discovery of a hereditary condition that affects the brain and spinal cord. The electromyography and MRI tests indicated deficiencies in the spinal cord and cerebellum, both of which are responsible for controlling coordinated movements. To what functional division of the nervous system would these structures belong?

EVERYDAY CONNECTION

How Much of Your Brain Do You Use?

Have you ever heard the claim that humans only use 10 percent of their brains? Maybe you have seen an advertisement on a website saying that there is a secret to unlocking the full potential of your mind—as if there were 90 percent of your brain sitting idle, just waiting for you to use it. If you see an ad like that, don't click. It isn't true.

An easy way to see how much of the brain a person uses is to take measurements of brain activity while performing a task. An example of this kind of measurement is functional magnetic resonance imaging (fMRI), which generates a map of the most active areas and can be generated and presented in three dimensions (Figure 12.7). This procedure is different from the standard MRI technique because it is measuring changes in the tissue in time with an experimental condition or event.

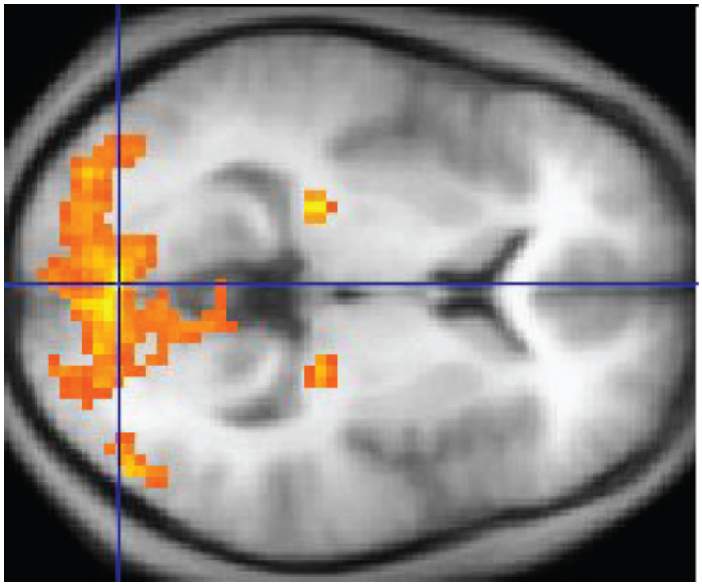


FIGURE 12.7 **fMRI** This fMRI shows activation of the visual cortex in response to visual stimuli. (credit: "Superborsuk"/Wikimedia Commons)

The underlying assumption is that active nervous tissue will have greater blood flow. By having the subject perform a visual task, activity all over the brain can be measured. Consider this possible experiment: the subject is told to look at a screen with a black dot in the middle (a fixation point). A photograph of a face is projected on the screen away from the center. The subject has to look at the photograph and decipher what it is. The subject has been instructed to push a button if the photograph is of someone they recognize. The photograph might be of a celebrity, so the subject would press the button, or it might be of a random person unknown to the subject, so the subject would not press the button.

In this task, visual sensory areas would be active, integrating areas would be active, motor areas responsible for moving the eyes would be active, and motor areas for pressing the button with a finger would be active. Those areas are distributed all around the brain and the fMRI images would show activity in more than just 10 percent of the brain (some evidence suggests that about 80 percent of the brain is using energy—based on blood flow to the tissue—during well-defined tasks similar to the one suggested above). This task does not even include all of the functions the brain performs. There is no language response, the body is mostly lying still in the MRI machine, and it does not consider the autonomic functions that would be ongoing in the background.

12.2 Nervous Tissue

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the basic structure of a neuron
- Identify the different types of neurons on the basis of polarity
- List the glial cells of the CNS and describe their function
- List the glial cells of the PNS and describe their function

Nervous tissue is composed of two types of cells, neurons and glial cells. Neurons are the primary type of cell that most anyone associates with the nervous system. They are responsible for the computation and communication that the nervous system provides. They are electrically active and release chemical signals to target cells. Glial cells, or glia, are known to play a supporting role for nervous tissue. Ongoing research pursues an expanded role that glial cells might play in signaling, but neurons are still considered the basis of this function. Neurons are important, but without glial support they would not be able to perform their function.

Neurons

Neurons are the cells considered to be the basis of nervous tissue. They are responsible for the electrical signals that communicate information about sensations, and that produce movements in response to those stimuli, along with inducing thought processes within the brain. An important part of the function of neurons is in their structure, or shape. The three-dimensional shape of these cells makes the immense numbers of connections within the nervous system possible.

Parts of a Neuron

As you learned in the first section, the main part of a neuron is the cell body, which is also known as the soma (soma = "body"). The cell body contains the nucleus and most of the major organelles. But what makes neurons special is that they have many extensions of their cell membranes, which are generally referred to as processes. Neurons are usually described as having one, and only one, axon—a fiber that emerges from the cell body and projects to target cells. That single axon can branch repeatedly to communicate with many target cells. It is the axon that propagates the nerve impulse, which is communicated to one or more cells. The other processes of the neuron are dendrites, which receive information from other neurons at specialized areas of contact called **synapses**. The dendrites are usually highly branched processes, providing locations for other neurons to communicate with the cell body. Information flows through a neuron from the dendrites, across the cell body, and down the axon. This gives the neuron a polarity—meaning that information flows in this one direction. Figure 12.8 shows the relationship of these parts to one another.

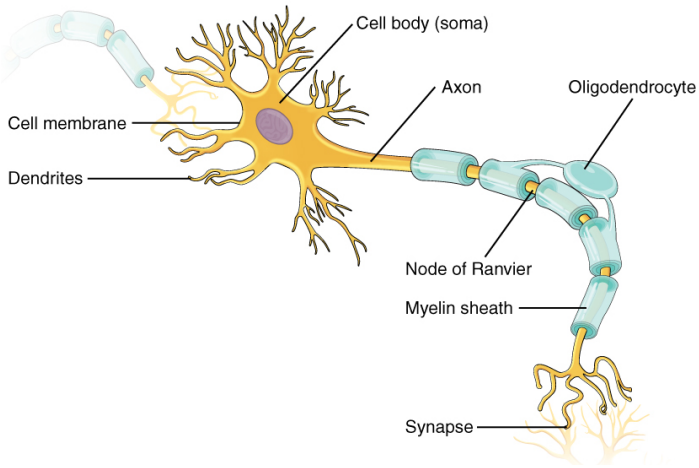


FIGURE 12.8 Parts of a Neuron The major parts of the neuron are labeled on a multipolar neuron from the CNS.

Where the axon emerges from the cell body, there is a special region referred to as the **axon hillock**. This is a tapering of the cell body toward the axon fiber. Within the axon hillock, the cytoplasm changes to a solution of limited components called **axoplasm**. As the axon hillock narrows, it transitions into the beginning of the axon called the initial segment. Action potentials are generated in the trigger zone, which is a combination of the axon hillock and **initial segment**.

Many axons are wrapped by an insulating substance called myelin, which is actually made from glial cells. Myelin acts as insulation much like the plastic or rubber that is used to insulate electrical wires. A key difference between myelin and the insulation on a wire is that there are gaps in the myelin covering of an axon. Each gap is called a **node of Ranvier** and is important to the way that electrical signals travel down the axon. The length of the axon between each gap, which is wrapped in myelin, is referred to as an **axon segment**. At the end of the axon is the **axon terminal**, where there are usually several branches extending toward the target cell, each of which ends in an enlargement called a **synaptic end bulb**. These bulbs are what make the connection with the target cell at the synapse.

Types of Neurons

There are many neurons in the nervous system—a number in the trillions. And there are many different types of neurons. They can be classified by many different criteria. The first way to classify them is by the number of

processes attached to the cell body. Using the standard model of neurons, one of these processes is the axon, and the rest are dendrites. Because information flows through the neuron from dendrites or cell bodies toward the axon, these names are based on the neuron's polarity (Figure 12.9).

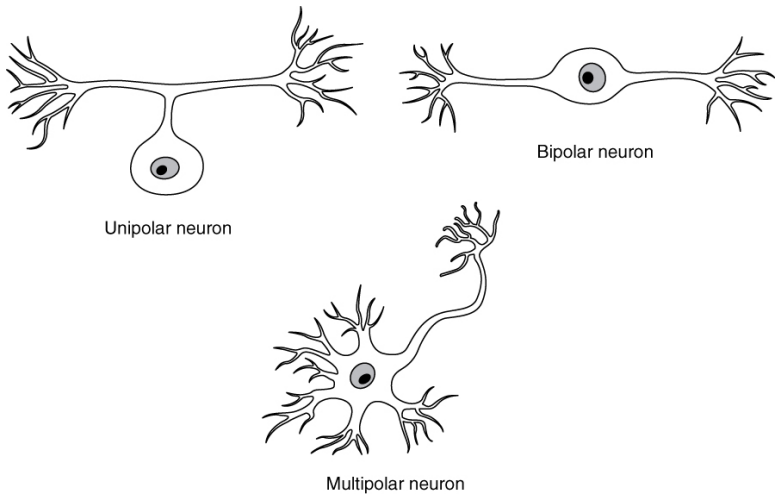


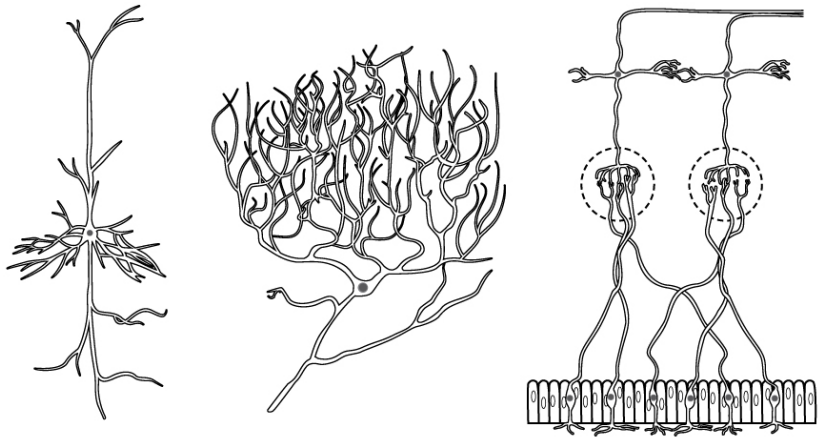
FIGURE 12.9 Neuron Classification by Shape Unipolar cells have one process that includes both the axon and dendrite. Bipolar cells have two processes, the axon and a dendrite. Multipolar cells have more than two processes, the axon and two or more dendrites.

Unipolar cells have only one process emerging from the cell. True unipolar cells are only found in invertebrate animals, so the unipolar cells in humans are more appropriately called "pseudo-unipolar" cells. Invertebrate unipolar cells do not have dendrites. Human unipolar cells have an axon that emerges from the cell body, but it splits so that the axon can extend along a very long distance. At one end of the axon are dendrites, and at the other end, the axon forms synaptic connections with a target. Unipolar cells are exclusively sensory neurons and have two unique characteristics. First, their dendrites are receiving sensory information, sometimes directly from the stimulus itself. Secondly, the cell bodies of unipolar neurons are always found in ganglia. Sensory reception is a peripheral function (those dendrites are in the periphery, perhaps in the skin) so the cell body is in the periphery, though closer to the CNS in a ganglion. The axon projects from the dendrite endings, past the cell body in a ganglion, and into the central nervous system.

Bipolar cells have two processes, which extend from each end of the cell body, opposite to each other. One is the axon and one the dendrite. Bipolar cells are not very common. They are found mainly in the olfactory epithelium (where smell stimuli are sensed), and as part of the retina.

Multipolar neurons are all of the neurons that are not unipolar or bipolar. They have one axon and two or more dendrites (usually many more). With the exception of the unipolar sensory ganglion cells, and the two specific bipolar cells mentioned above, all other neurons are multipolar. Some cutting edge research suggests that certain neurons in the CNS do not conform to the standard model of "one, and only one" axon. Some sources describe a fourth type of neuron, called an anaxonic neuron. The name suggests that it has no axon (an- = "without"), but this is not accurate. Anaxonic neurons are very small, and if you look through a microscope at the standard resolution used in histology (approximately 400X to 1000X total magnification), you will not be able to distinguish any process specifically as an axon or a dendrite. Any of those processes can function as an axon depending on the conditions at any given time. Nevertheless, even if they cannot be easily seen, and one specific process is definitively the axon, these neurons have multiple processes and are therefore multipolar.

Neurons can also be classified on the basis of where they are found, who found them, what they do, or even what chemicals they use to communicate with each other. Some neurons referred to in this section on the nervous system are named on the basis of those sorts of classifications (Figure 12.10). For example, a multipolar neuron that has a very important role to play in a part of the brain called the cerebellum is known as a Purkinje (commonly pronounced per-KIN-gee) cell. It is named after the anatomist who discovered it (Jan Evangelista Purkinje, 1787–1869).



(a) Pyramidal cell of the cerebral cortex (b) Purkinje cell of the cerebellar cortex (c) Olfactory cells in the olfactory epithelium and olfactory bulbs

FIGURE 12.10 Other Neuron Classifications Three examples of neurons that are classified on the basis of other criteria. (a) The pyramidal cell is a multipolar cell with a cell body that is shaped something like a pyramid. (b) The Purkinje cell in the cerebellum was named after the scientist who originally described it. (c) Olfactory neurons are named for the functional group with which they belong.

Glial Cells

Glial cells, or neuroglia or simply glia, are the other type of cell found in nervous tissue. They are considered to be supporting cells, and many functions are directed at helping neurons complete their function for communication. The name glia comes from the Greek word that means "glue," and was coined by the German pathologist Rudolph Virchow, who wrote in 1856: "This connective substance, which is in the brain, the spinal cord, and the special sense nerves, is a kind of glue (neuroglia) in which the nervous elements are planted." Today, research into nervous tissue has shown that there are many deeper roles that these cells play. And research may find much more about them in the future.

There are six types of glial cells. Four of them are found in the CNS and two are found in the PNS. Table 12.2 outlines some common characteristics and functions.

Glial Cell Types by Location and Basic Function

CNS glia	PNS glia	Basic function
Astrocyte	Satellite cell	Support
Oligodendrocyte	Schwann cell	Insulation, myelination
Microglia	-	Immune surveillance and phagocytosis
Ependymal cell	-	Creating CSF

TABLE 12.2

Glial Cells of the CNS

One cell providing support to neurons of the CNS is the **astrocyte**, so named because it appears to be star-shaped under the microscope (astro- = "star"). Astrocytes have many processes extending from their main cell body (not axons or dendrites like neurons, just cell extensions). Those processes extend to interact with neurons, blood vessels, or the connective tissue covering the CNS that is called the pia mater (Figure 12.11). Generally, they are supporting cells for the neurons in the central nervous system. Some ways in which they support neurons in the central nervous system are by maintaining the concentration of chemicals in the extracellular space, removing excess signaling molecules, reacting to tissue damage, and contributing to the **blood-brain barrier (BBB)**. The

blood-brain barrier is a physiological barrier that keeps many substances that circulate in the rest of the body from getting into the central nervous system, restricting what can cross from circulating blood into the CNS. Nutrient molecules, such as glucose or amino acids, can pass through the BBB, but other molecules cannot. This actually causes problems with drug delivery to the CNS. Pharmaceutical companies are challenged to design drugs that can cross the BBB as well as have an effect on the nervous system.

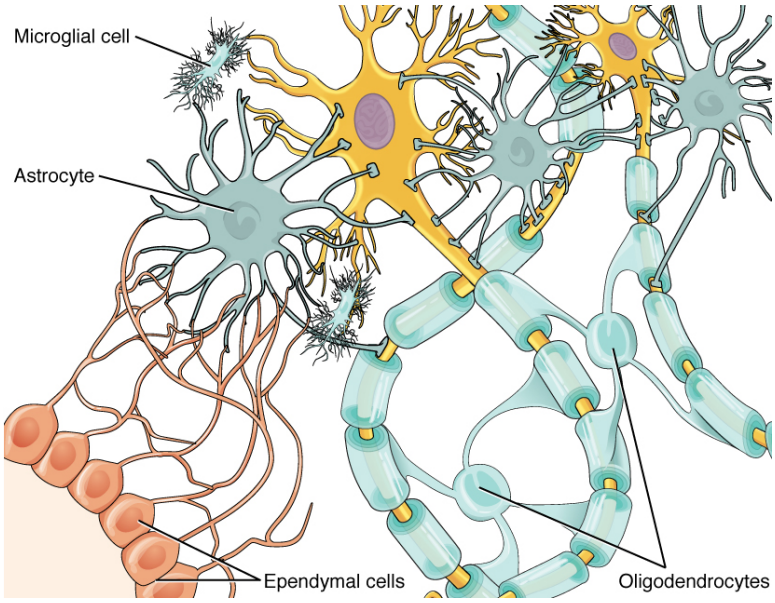


FIGURE 12.11 Glial Cells of the CNS The CNS has astrocytes, oligodendrocytes, microglia, and ependymal cells that support the neurons of the CNS in several ways.

Like a few other parts of the body, the brain has a privileged blood supply. Very little can pass through by diffusion. Most substances that cross the wall of a blood vessel into the CNS must do so through an active transport process. Because of this, only specific types of molecules can enter the CNS. Glucose—the primary energy source—is allowed, as are amino acids. Water and some other small particles, like gases and ions, can enter. But most everything else cannot, including white blood cells, which are one of the body's main lines of defense. While this barrier protects the CNS from exposure to toxic or pathogenic substances, it also keeps out the cells that could protect the brain and spinal cord from disease and damage. The BBB also makes it harder for pharmaceuticals to be developed that can affect the nervous system. Aside from finding efficacious substances, the means of delivery is also crucial.

Also found in CNS tissue is the **oligodendrocyte**, sometimes called just "oligo," which is the glial cell type that insulates axons in the CNS. The name means "cell of a few branches" (oligo- = "few"; dendro- = "branches"; -cyte = "cell"). There are a few processes that extend from the cell body. Each one reaches out and surrounds an axon to insulate it in myelin. One oligodendrocyte will provide the myelin for multiple axon segments, either for the same axon or for separate axons. The function of myelin will be discussed below.

Microglia are, as the name implies, smaller than most of the other glial cells. Ongoing research into these cells, although not entirely conclusive, suggests that they may originate as white blood cells, called macrophages, that become part of the CNS during early development. While their origin is not conclusively determined, their function is related to what macrophages do in the rest of the body. When macrophages encounter diseased or damaged cells in the rest of the body, they ingest and digest those cells or the pathogens that cause disease. Microglia are the cells in the CNS that can do this in normal, healthy tissue, and they are therefore also referred to as CNS-resident macrophages.

The **ependymal cell** is a glial cell that filters blood to make **cerebrospinal fluid (CSF)**, the fluid that circulates through the CNS. Because of the privileged blood supply inherent in the BBB, the extracellular space in nervous tissue does not easily exchange components with the blood. Ependymal cells line each **ventricle**, one of four central cavities that are remnants of the hollow center of the neural tube formed during the embryonic development of the

brain. The **choroid plexus** is a specialized structure in the ventricles where ependymal cells come in contact with blood vessels and filter and absorb components of the blood to produce cerebrospinal fluid. Because of this, ependymal cells can be considered a component of the BBB, or a place where the BBB breaks down. These glial cells appear similar to epithelial cells, making a single layer of cells with little intracellular space and tight connections between adjacent cells. They also have cilia on their apical surface to help move the CSF through the ventricular space. The relationship of these glial cells to the structure of the CNS is seen in Figure 12.11.

Glial Cells of the PNS

One of the two types of glial cells found in the PNS is the **satellite cell**. Satellite cells are found in sensory and autonomic ganglia, where they surround the cell bodies of neurons. This accounts for the name, based on their appearance under the microscope. They provide support, performing similar functions in the periphery as astrocytes do in the CNS—except, of course, for establishing the BBB.

The second type of glial cell is the **Schwann cell**, which insulate axons with myelin in the periphery. Schwann cells are different than oligodendrocytes, in that a Schwann cell wraps around a portion of only one axon segment and no others. Oligodendrocytes have processes that reach out to multiple axon segments, whereas the entire Schwann cell surrounds just one axon segment. The nucleus and cytoplasm of the Schwann cell are on the edge of the myelin sheath. The relationship of these two types of glial cells to ganglia and nerves in the PNS is seen in Figure 12.12.

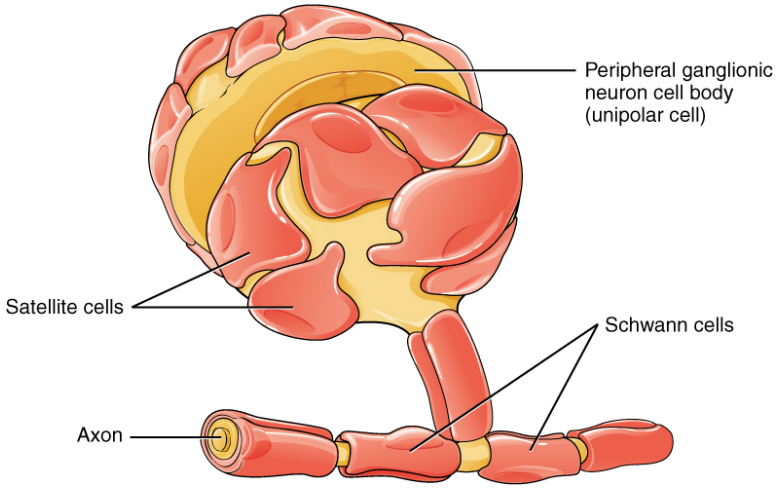


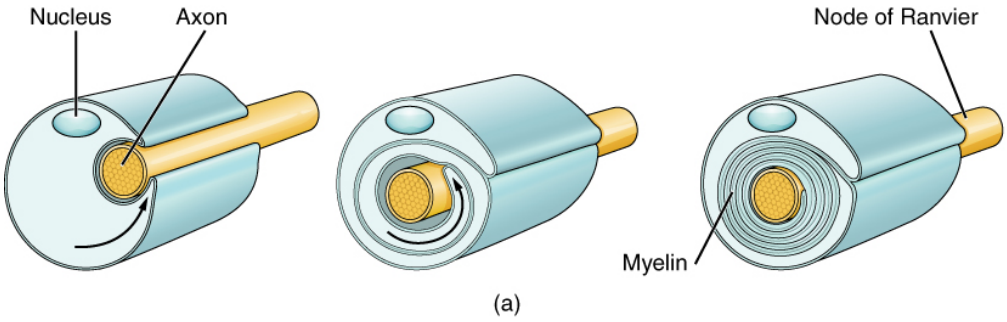
FIGURE 12.12 Glial Cells of the PNS The PNS has satellite cells and Schwann cells.

Myelin

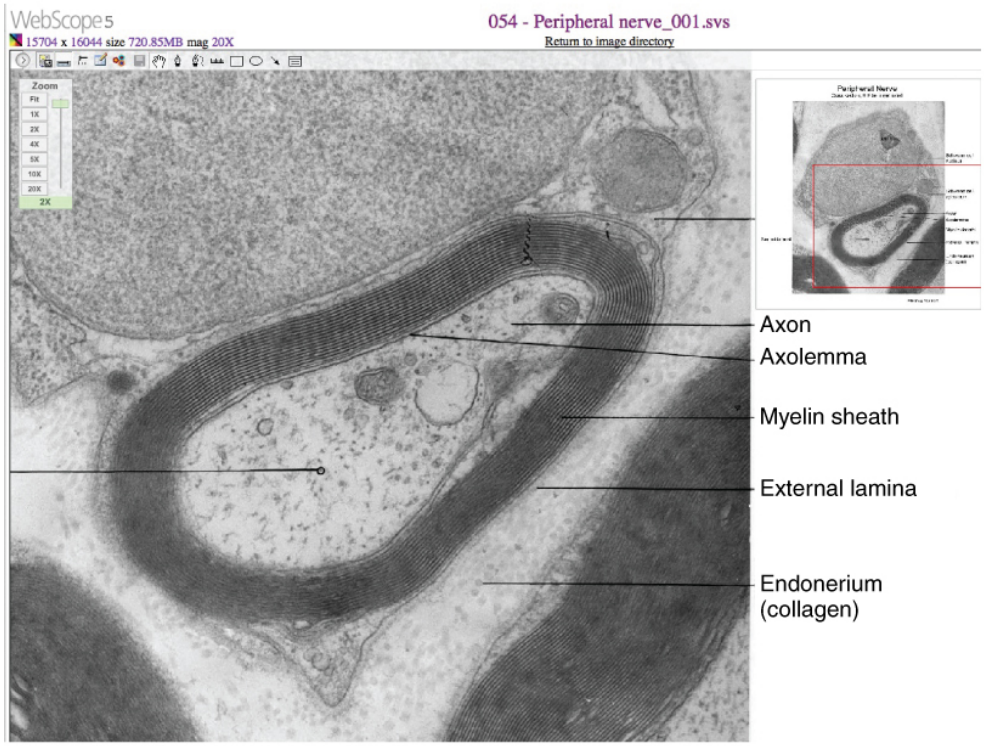
The insulation for axons in the nervous system is provided by glial cells, oligodendrocytes in the CNS, and Schwann cells in the PNS. Whereas the manner in which either cell is associated with the axon segment, or segments, that it insulates is different, the means of myelinating an axon segment is mostly the same in the two situations. Myelin is a lipid-rich sheath that surrounds the axon and by doing so creates a **myelin sheath** that facilitates the transmission of electrical signals along the axon. The lipids are essentially the phospholipids of the glial cell membrane. Myelin, however, is more than just the membrane of the glial cell. It also includes important proteins that are integral to that membrane. Some of the proteins help to hold the layers of the glial cell membrane closely together.

The appearance of the myelin sheath can be thought of as similar to the pastry wrapped around a hot dog for “pigs in a blanket” or a similar food. The glial cell is wrapped around the axon several times with little to no cytoplasm between the glial cell layers. For oligodendrocytes, the rest of the cell is separate from the myelin sheath as a cell process extends back toward the cell body. A few other processes provide the same insulation for other axon segments in the area. For Schwann cells, the outermost layer of the cell membrane contains cytoplasm and the nucleus of the cell as a bulge on one side of the myelin sheath. During development, the glial cell is loosely or incompletely wrapped around the axon (Figure 12.13**a**). The edges of this loose enclosure extend toward each other, and one end tucks under the other. The inner edge wraps around the axon, creating several layers, and the other edge closes around the outside so that the axon is completely enclosed.

Myelin sheaths can extend for one or two millimeters, depending on the diameter of the axon. Axon diameters can be as small as 1 to 20 micrometers. Because a micrometer is 1/1000 of a millimeter, this means that the length of a myelin sheath can be 100–1000 times the diameter of the axon. Figure 12.8, Figure 12.11, and Figure 12.12 show the myelin sheath surrounding an axon segment, but are not to scale. If the myelin sheath were drawn to scale, the neuron would have to be immense—possibly covering an entire wall of the room in which you are sitting.



(a)



(b)

FIGURE 12.13 The Process of Myelination Myelinating glia wrap several layers of cell membrane around the cell membrane of an axon segment. A single Schwann cell insulates a segment of a peripheral nerve, whereas in the CNS, an oligodendrocyte may provide insulation for a few separate axon segments. EM × 1,460,000. (Micrograph provided by the Regents of University of Michigan Medical School © 2012)

DISORDERS OF THE...

Nervous Tissue

Several diseases can result from the demyelination of axons. The causes of these diseases are not the same; some have genetic causes, some are caused by pathogens, and others are the result of autoimmune disorders. Though the causes are varied, the results are largely similar. The myelin insulation of axons is compromised,

making electrical signaling slower.

Multiple sclerosis (MS) is one such disease. It is an example of an autoimmune disease. The antibodies produced by lymphocytes (a type of white blood cell) mark myelin as something that should not be in the body. This causes inflammation and the destruction of the myelin in the central nervous system. As the insulation around the axons is destroyed by the disease, scarring becomes obvious. This is where the name of the disease comes from; sclerosis means hardening of tissue, which is what a scar is. Multiple scars are found in the white matter of the brain and spinal cord. The symptoms of MS include both somatic and autonomic deficits. Control of the musculature is compromised, as is control of organs such as the bladder.

Guillain-Barré (pronounced gee-YAN bah-RAY) syndrome is an example of a demyelinating disease of the peripheral nervous system. It is also the result of an autoimmune reaction, but the inflammation is in peripheral nerves. Sensory symptoms or motor deficits are common, and autonomic failures can lead to changes in the heart rhythm or a drop in blood pressure, especially when standing, which causes dizziness.

12.3 The Function of Nervous Tissue

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Distinguish the major functions of the nervous system: sensation, integration, and response
- List the sequence of events in a simple sensory receptor–motor response pathway

Having looked at the components of nervous tissue, and the basic anatomy of the nervous system, next comes an understanding of how nervous tissue is capable of communicating within the nervous system. Before getting to the nuts and bolts of how this works, an illustration of how the components come together will be helpful. An example is summarized in Figure 12.14.

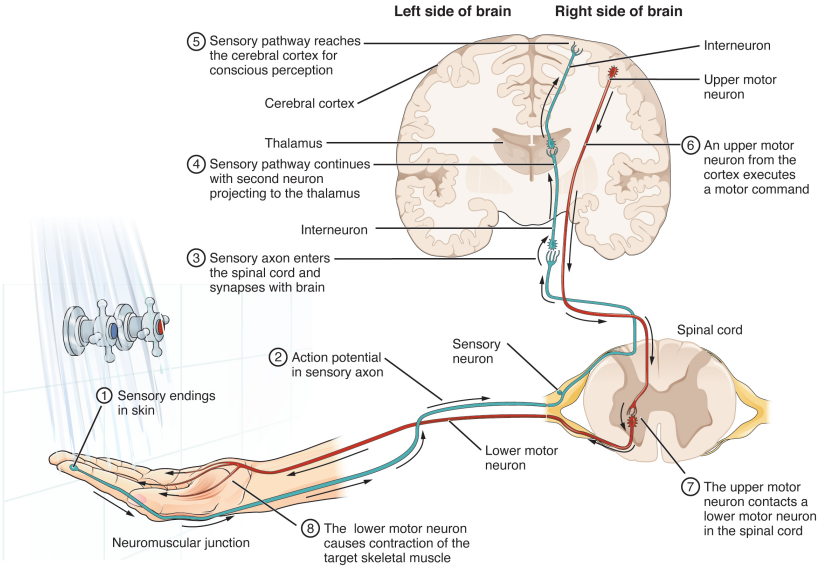


FIGURE 12.14 Testing the Water: Posterior View (1) The sensory neuron has endings in the skin that sense a stimulus such as water temperature. The strength of the signal that starts here is dependent on the strength of the stimulus. (2) The graded potential from the sensory endings, if strong enough, will initiate an action potential at the initial segment of the axon (which is immediately adjacent to the sensory endings in the skin). (3) The axon of the peripheral sensory neuron enters the spinal cord and contacts another neuron in the gray matter. The contact is a synapse where another graded potential is caused by the release of a chemical signal from the axon terminals. (4) An action potential is initiated at the initial segment of this neuron and travels up the sensory pathway to a region of the brain called the thalamus. Another synapse passes the information along to the next neuron. (5) The sensory pathway ends when the signal reaches the cerebral cortex. (6) After integration with neurons in other parts of the cerebral cortex, a motor command is sent from the precentral gyrus of the frontal cortex. (7) The upper motor neuron sends an action potential down to the spinal cord. The target of the upper motor neuron is the dendrites of the lower motor neuron in the gray matter of the spinal cord. (8) The axon of the lower motor neuron emerges from the spinal cord in a nerve and connects to a muscle through a neuromuscular junction to cause contraction of the target muscle.

Imagine you are about to take a shower in the morning before going to school. You have turned on the faucet to start the water as you prepare to get in the shower. After a few minutes, you expect the water to be a temperature that will be comfortable to enter. So you put your hand out into the spray of water. What happens next depends on how your nervous system interacts with the stimulus of the water temperature and what you do in response to that stimulus.

Found in the skin of your fingers or toes is a type of sensory receptor that is sensitive to temperature, called a **thermoreceptor**. When you place your hand under the shower (Figure 12.15), the cell membrane of the thermoreceptors changes its electrical state (voltage). The amount of change is dependent on the strength of the stimulus (how hot the water is). This is called a **graded potential**. If the stimulus is strong, the voltage of the cell membrane will change enough to generate an electrical signal that will travel down the axon. You have learned about this type of signaling before, with respect to the interaction of nerves and muscles at the neuromuscular junction. The voltage at which such a signal is generated is called the **threshold**, and the resulting electrical signal is called an **action potential**. In this example, the action potential travels—a process known as **propagation**—along the axon from the axon hillock to the axon terminals and into the synaptic end bulbs. When this signal reaches the end bulbs, it causes the release of a signaling molecule called a **neurotransmitter**.

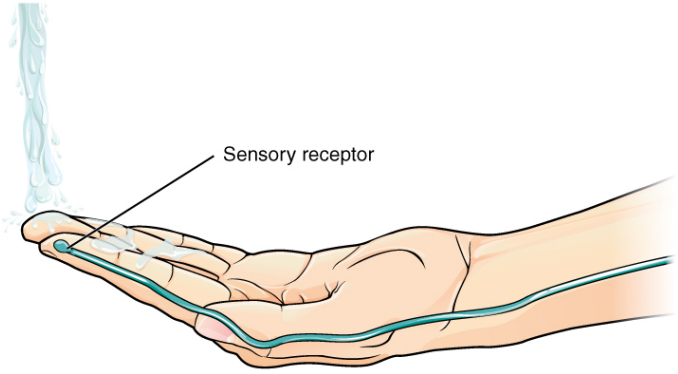


FIGURE 12.15 The Sensory Input Receptors in the skin sense the temperature of the water.

The neurotransmitter diffuses across the short distance of the synapse and binds to a receptor protein of the target neuron. When the molecular signal binds to the receptor, the cell membrane of the target neuron changes its electrical state and a new graded potential begins. If that graded potential is strong enough to reach threshold, the second neuron generates an action potential at its axon hillock. The target of this neuron is another neuron in the **thalamus** of the brain, the part of the CNS that acts as a relay for sensory information. At another synapse, neurotransmitter is released and binds to its receptor. The thalamus then sends the sensory information to the **cerebral cortex**, the outermost layer of gray matter in the brain, where conscious perception of that water temperature begins.

Within the cerebral cortex, information is processed among many neurons, integrating the stimulus of the water temperature with other sensory stimuli, with your emotional state (you just aren't ready to wake up; the bed is calling to you), memories (perhaps of the lab notes you have to study before a quiz). Finally, a plan is developed about what to do, whether that is to turn the temperature up, turn the whole shower off and go back to bed, or step into the shower. To do any of these things, the cerebral cortex has to send a command out to your body to move muscles (Figure 12.16).